

Practical - 1

Practical Title: Project Initiation, Requirement Analysis, Sitemap, Wireframe, and GitHub Setup

Problem Definition: Initiate the semester-long StudentHub portal by identifying the problem scope, user roles, key modules, navigation flow, and minimum 10 pages. Create a sitemap, low-fidelity wireframe, project folder structure, README file, and GitHub repository.

A. Theory

1. What is URL and parts of URL?

Answer:

A URL (Uniform Resource Locator) is the address of a web page or any resource available on the Internet. It helps the browser find and open a specific website or file.

A URL has the following parts:

- Protocol: Shows how data is transferred (HTTP or HTTPS).
- Domain Name: The name of the website (for example, www.google.com).
- Path: Shows the location of a page or file on the website.
- Query Parameters: Used to send extra information to the server.
- Fragment: Takes the user to a specific section of a webpage.

Example:

`https://www.studenthub.com/profile?id=101#details`

- Protocol → `https`
- Domain Name → `www.studenthub.com`
- Path → `/profile`
- Query Parameter → `id=101`
- Fragment → `details`

Reference:

- <https://www.w3schools.com/html>

2. How does HTML file process in web browser?

Answer:

When we open an HTML file in a web browser, the browser first reads the HTML code. It understands the HTML tags and creates the structure of the webpage.

If the HTML file contains CSS, the browser applies styles to the page. If JavaScript is included, the browser runs the script to make the page interactive.

Finally, the browser displays the complete webpage on the screen.

Process:

1. Browser reads the HTML file.
2. HTML is parsed.
3. DOM (Document Object Model) is created.
4. CSS styles are applied.
5. JavaScript is executed.
6. The webpage is displayed.

Reference:

- <https://www.w3schools.com/html>

3. How will page navigation flow be managed among all HTML pages?

Answer:

In the StudentHub project, page navigation will be managed using navigation links created with the HTML `<a>` (anchor) tag. Every page will have the same navigation menu so users can easily move from one page to another.

The navigation menu may include:

- Home
- Login
- Dashboard
- Courses
- Assignments
- Attendance
- Results
- Profile
- Feedback
- Contact

This navigation system makes the website easy to use and improves the user experience.

Reference:

- https://www.w3schools.com/html/html_links.asp
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4. How will GitHub commits be maintained after each practical?

Answer:

After completing each practical, the updated files will be saved and uploaded to GitHub using a new commit. Every commit will have a clear and meaningful message so that the changes can be identified easily.

The basic steps are:

1. Save all project files.
2. Run `git add .`
3. Run `git commit -m "Completed Practical 1"`
4. Run `git push origin main`

This process keeps a proper history of the project. It also helps us track changes, fix mistakes, and manage different versions of the project.

Reference:

<https://git-scm.com/docs>

B) Implementation/ Output/Result :

functional Requirements:

Project Name - studenthub portal

Objective:

StudentHub Portal is a web application designed to help students manage their academic activities efficiently.

User Roles:

(1) Student:

Register Account

Login

View Dashboard

View Courses

View Attendance

View Results

Submit Feedback

Update Profile

(2)Admin:

Manage Students

Manage Courses

Manage Attendance

Manage Results

View Student Feedback

Functional Modules:

1. Home

Welcome Page

Navigation Bar

StudentHub Introduction

2. Assignment

View Assignment

Assignment Details

Submission Date

3. Attendance

View Attendance Percentage

Subject-wise Attendance

4. Contact

Contact Form

College Information

5. Course

View Available Courses

Course Description

6. Dashboard

Quick Overview

Navigation Cards

7. Login

Username

Password

Authentication

8. Profile

Student Details

Update Information

9. Register

Create New Account

10. Result

Subject Marks

Grade

CGPA

11. Student Feedback

Give Feedback

Rate College Services

Code structure:

Home (Dashboard)

|

|— About

└─ **Assignment**

| └─ **Assignment List**

| └─ **Upload Assignment**

|

└─ **Attendance**

| └─ **Attendance Record**

|

└─ **Course**

| └─ **Available Courses**

|

└─ **Contact**

| └─ **Contact Form**

|

└─ **Login**

| └─ **Signup**

| └─ **Forgot Password**

|

└─ **Profile**

| └─ **Student Details**

|

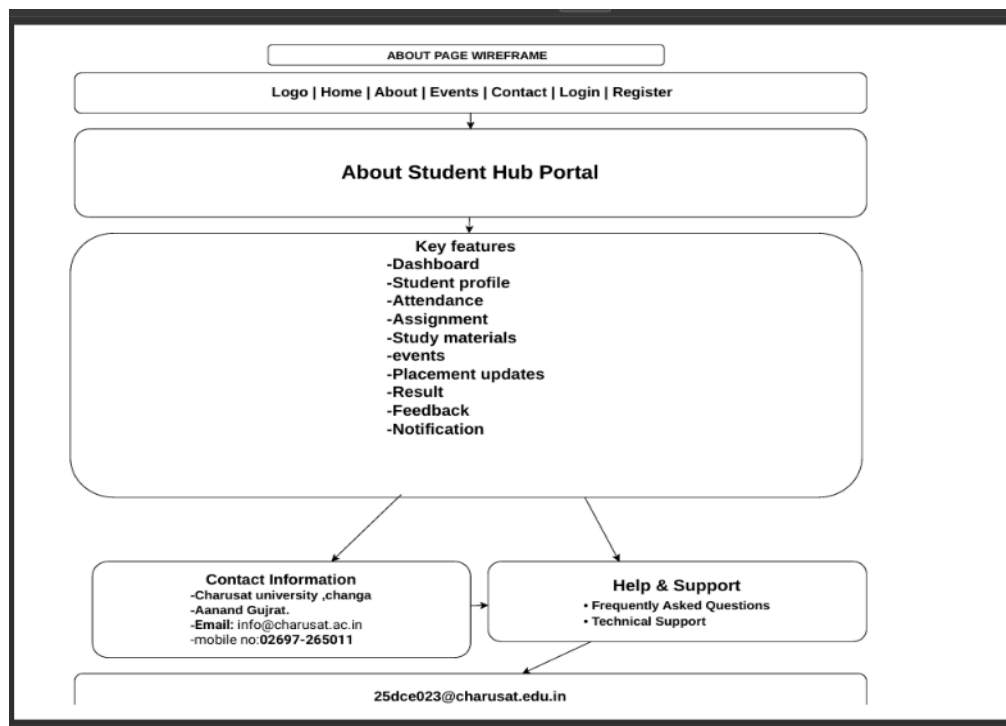
└─ **Result**

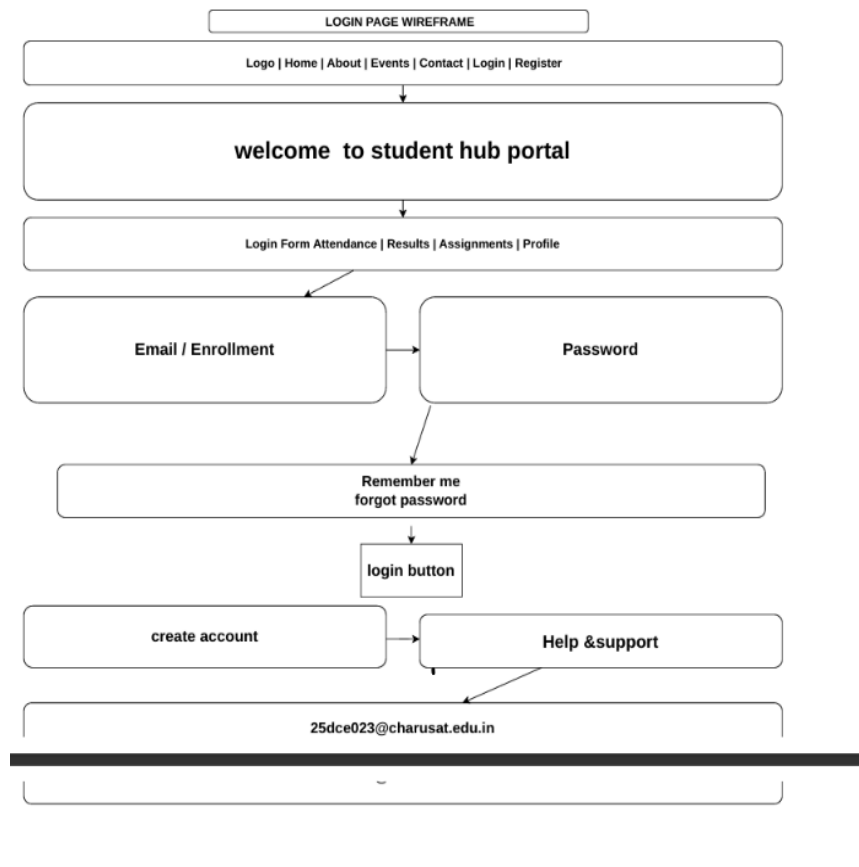
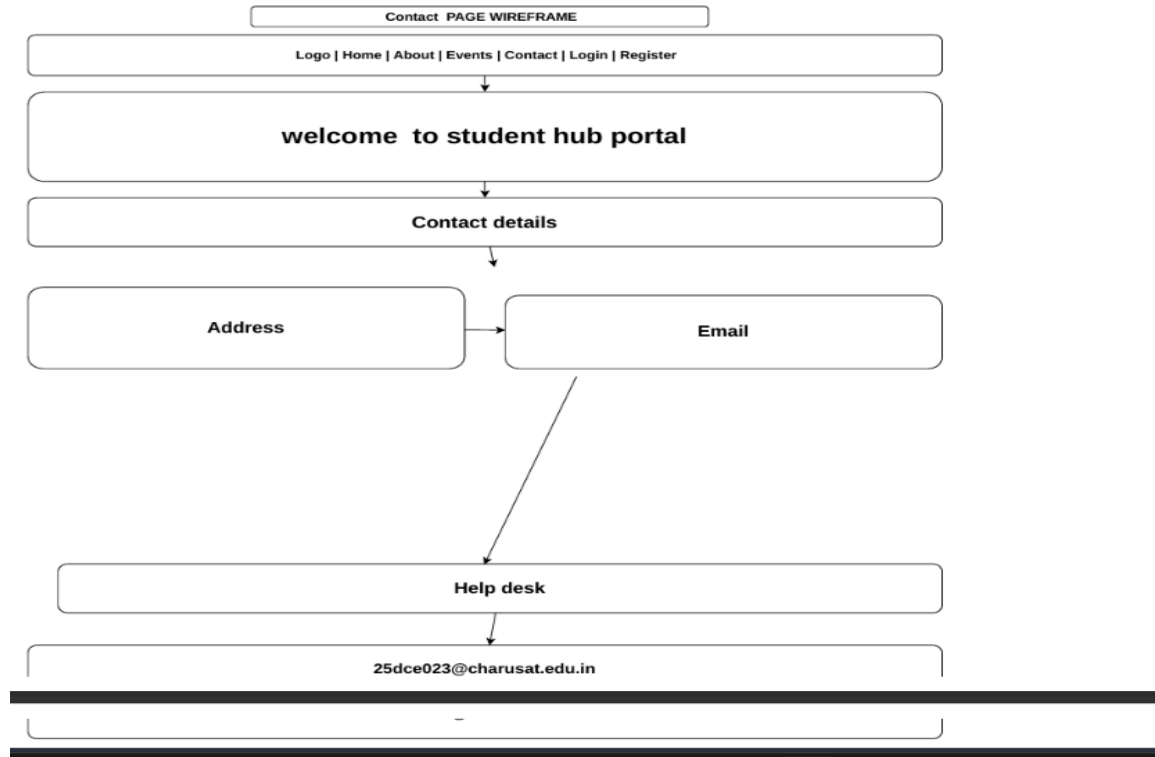
- | └─ Semester Marks
- |
- | └─ Faculty Feedback
- └─ Rating & Comments

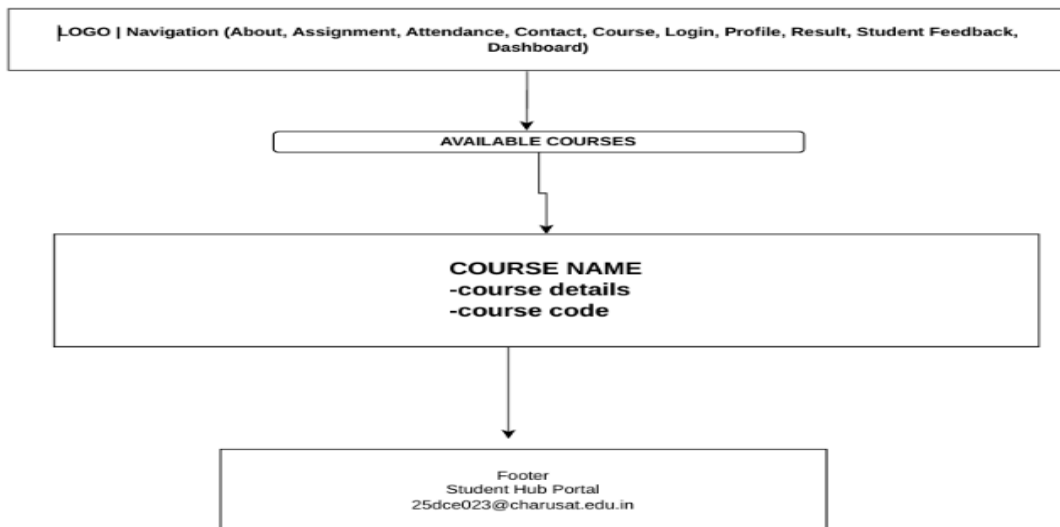
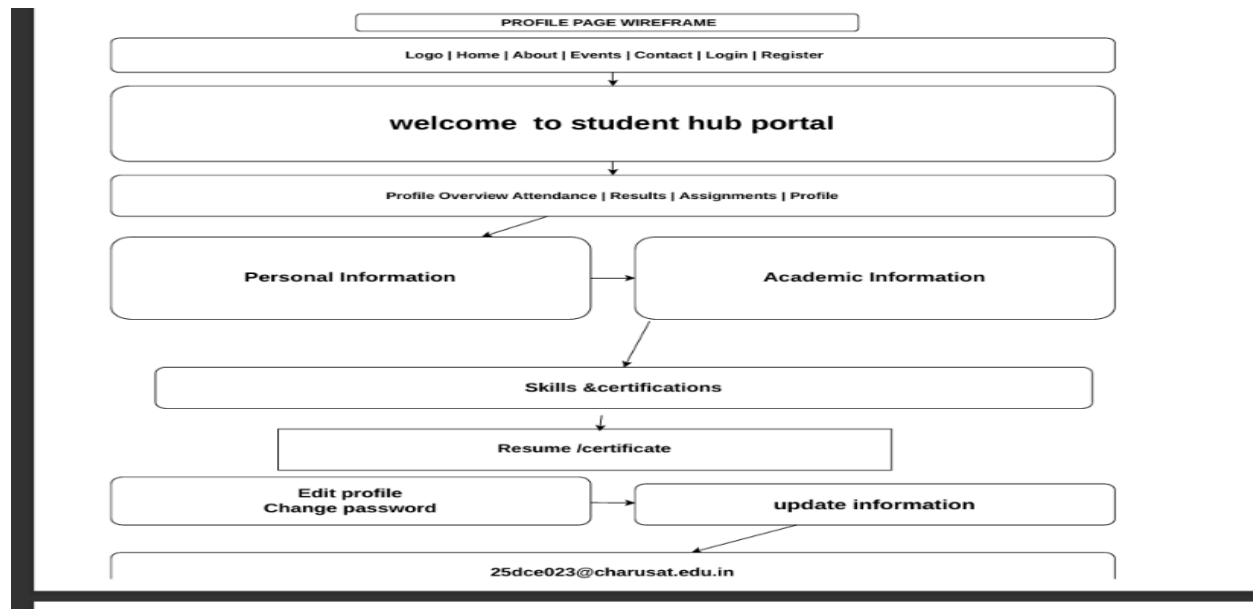
Tools Technologys : vs code, draw.io , Git, Github

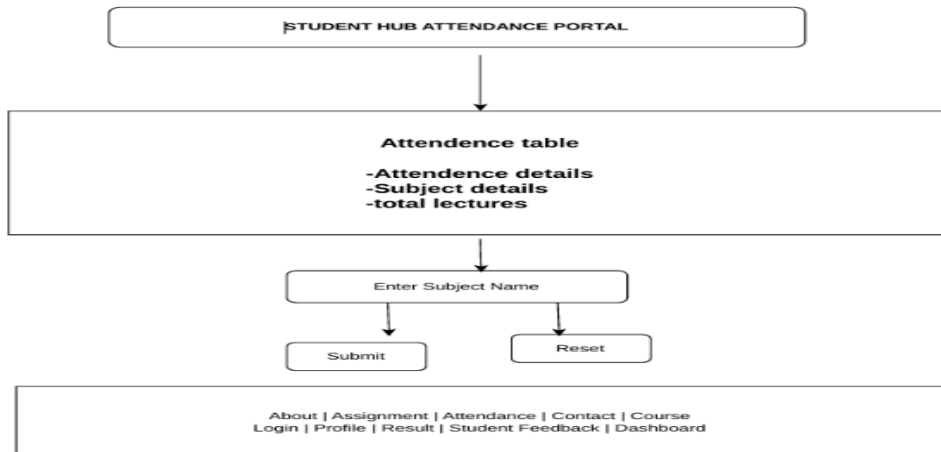
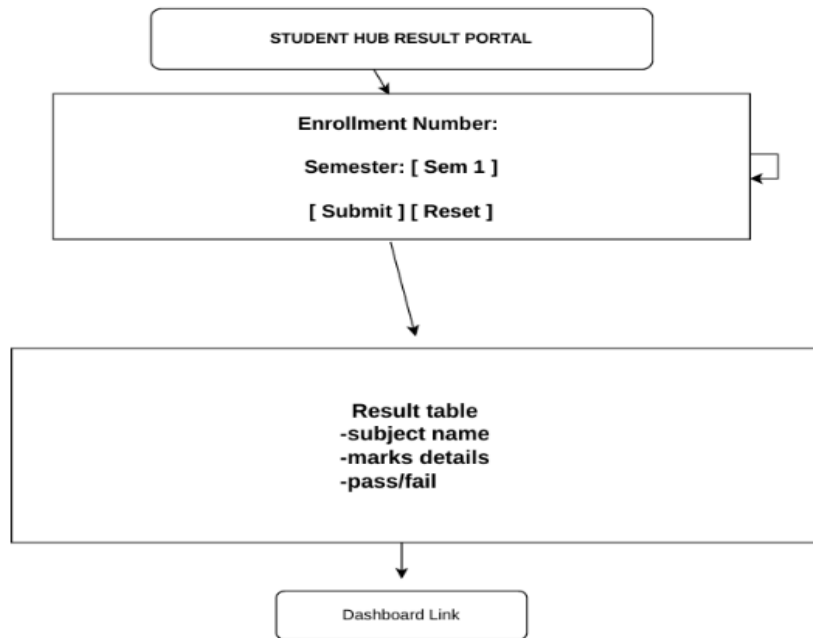
C) Evidence of work :

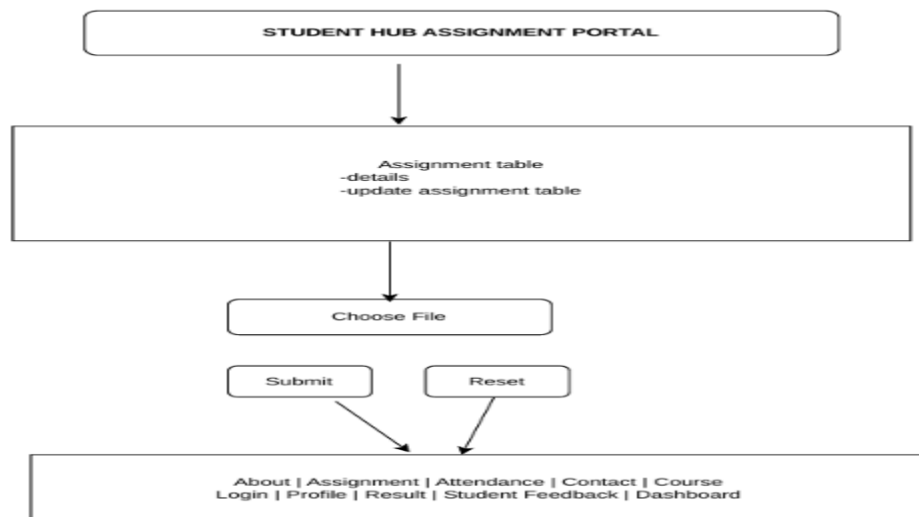
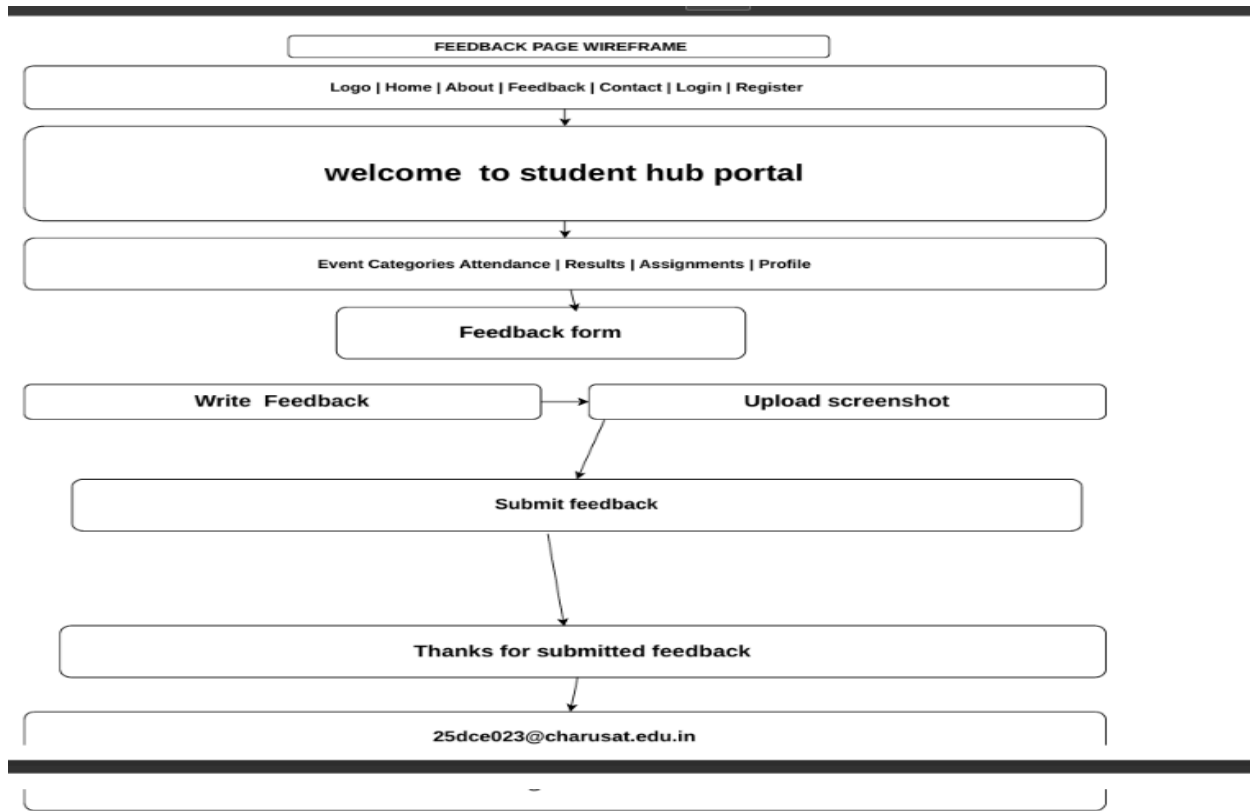
Wireframe screenshots:



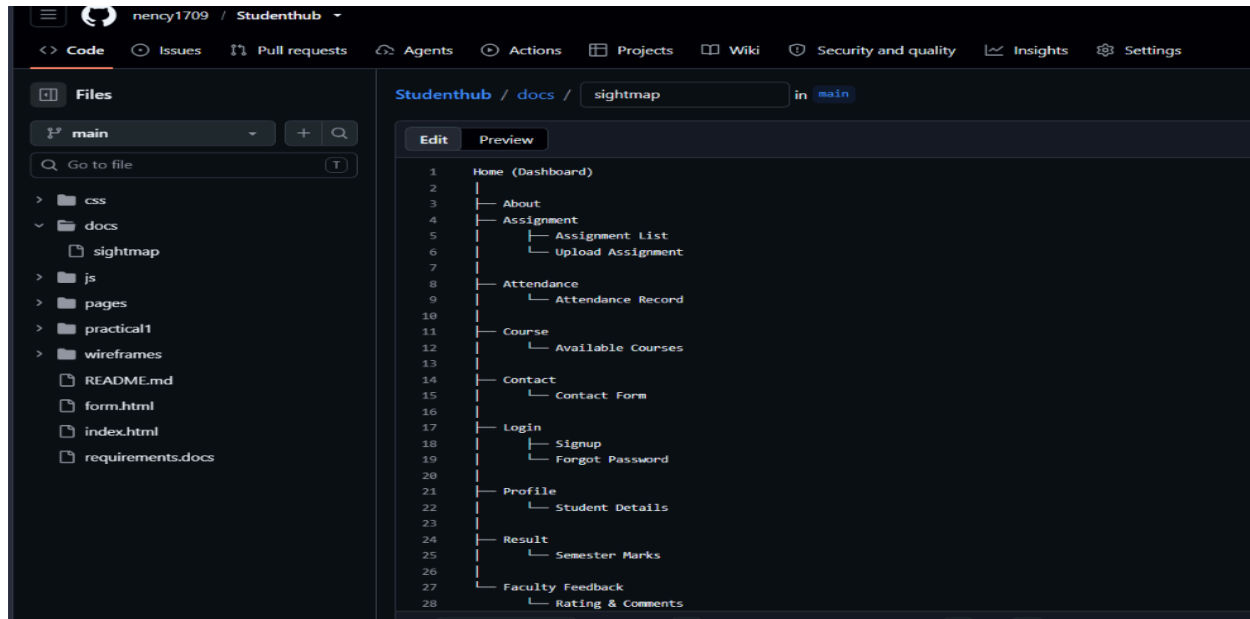








Sightmap screenshots:



D) Github repository link :

[nency1709/Studenthub: practical 1](#)

E) Problem facing :

Deciding the correct page hierarchy was difficult.

Managing navigation between all HTML pages was confusing.

Designing a clear wireframe layout took multiple attempts.

Understanding user flow required careful planning.

After a few modifications, the sitemap and wireframe became more organized

F) Learning outcome :

I learned how to plan a website before starting development.

I understood the importance of requirement analysis in a project.

I learned how to create a sitemap to organize website pages.

I learned to design a basic wireframe for planning the page layout.

- I understood how to manage project files and upload them to GitHub using commits.
- This practical improved my understanding of website structure and navigation.

G) Faculty remarks :